

Honeywell

Experion

**Planning, Installation, and Service
for PE1600/PE1800**

EP-DPCXX2

Release Independent

06/05

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About This Document

This document contains installation and service information for the PE1600 Honeywell computer platform.

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Planning, Installation, and Service for PE1600/PE1800 – EX16	EP-DPCXX2	Release Independent	06/05

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
TPS Users	
<i>TPS System Site Planning</i>	SW02-550 or later
<i>TPS System Implementation Guide for Windows 2000</i>	TP08W
<i>TPS System Planning Guide for Windows 2000</i>	TP10W
<i>TPS System Administration Guide for Windows 2000</i>	TP06W
Experion PKS Users	
<i>Experion PKS Overview</i>	EP-DCSX32 or later
<i>Experion PKS Software Installation and Upgrade Guide</i>	EP-DCXX12 or later
<i>Server and Client Planning Guide</i>	EP-DSX132 or later
<i>Server and Client Configuration Guide (for Experion PKS)</i>	EP-DSXX22 or later
<i>Experion PKS Operators Guide</i>	EP-DSXX42 or later
FTE Users	
<i>Fault Tolerant Ethernet Installation and Service Guide e</i>	FE05

1-800 343-0228

Contacts

Contacts

World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our Process Solutions customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Process Solutions	http://www.acs.honeywell.com
International	http://content.honeywell.com/global/

Telephone

Contact us by telephone at the numbers listed below.

Organization		Phone Number	
United States and Canada	Honeywell International Inc. Process Solutions	1-800-343-0228	Sales
		1-800-525-7439	Service
		1-800-822-7673	Tech. Support
Asia Pacific	Honeywell Asia Pacific Inc. Hong Kong	(852) 23 31 9133	
Europe	Honeywell PACE Brussels, Belgium	[32-2] 728-2711	
Latin America	Honeywell International Inc. Sunrise, Florida U.S.A.	(954) 845-2600	

Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the book set.
	REFERENCE - INTERNAL: Identifies an additional source of information within the book set.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.

Symbol	Definition
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices.
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. System Planning

1.1 Overview

About the PE1600/PE1800 platforms

The PE1600 and PE1800 platforms provide PC-based functionality for Honeywell systems. For TPS systems, the platforms have an LCNP4 card installed, which allows connection to the TotalPlant Process Network (TPN) coax through the LCN Media Access Unit (MAU). Additionally, the platforms connect to the Ethernet through a standard Ethernet adapter card, or the optional FTE Ethernet adapter card.

Software requirements

The PE1600 is qualified on the Windows 2000 server operating system. The PE1800 is qualified for on the Windows 2000 or 2003 server operating systems. Refer to the latest SCN for software applications that have been qualified for use on these platforms.

BIOS configuration

Each platform must have the following releases of BIOS installed:

- PE1600SC must have release A08 or a later version.
- PE1800 must have release A00 or a later version.

1.2 PE1600 Description

Enclosure

The PE1600 is the PC platform used for TPS and Experion PKS nodes, and can be mounted in cabinet furniture or used as a desktop.

When mounted in a cabinet, the enclosure uses the following amount of space:

- Fixed mount: 5U (8.75") of rack space
- Slide mount: 6U (10.5") of rack space

Electronics module for PE1600

The peripheral electronics assemblies for the PE1600 are based upon either the Peripheral Component Interconnect (PCI) bus, AGP or USB 1.1. The standard SDRAM memory for this platform is 512 MB (1 – 512 MB DDR SDRAM), optionally expandable to 2.0 GB, with no cache memory options.

Storage and media devices

The standard mass storage for the PE1600 platform is the 36 GB SCSI hard drive with a mirrored RAID-1 hard drive. The hard drive is connected to the RAID-1 SCSI controller card in the PCI1 slot.

All mass storage devices are connected via IDE or SCSI. The floppy drive connects to the floppy disk connector on the motherboard and the EIDE CDRW/DVD-ROM connects to the IDE connector, also on the motherboard.

High availability

The PE1600, Honeywell's high-availability server computer platform is intended to be used for non-redundant topology configurations. High-availability refers to the ability to provide some fail-safe characteristics through the use of RAID disk configurations with hot-swappable disk drives and redundant hot-swappable power supplies.

PE1600 common standard features

The following is a list of the common features of this platform:

- Dell motherboard with Intel Xeon single CPU
- 512 KB Cache
- Bus Type: two 32-bit 33-MHz PCIs
- Embedded dual channel ATA 100
- Ports: 1 serial, 1 parallel, 2 USB V1.1/2.0
- Integrated LAN (10/100/1000 Mb/Sec)
- Bus Type: PCI and PCI-X
- Expansion slots: two 32-bit, 33-MHz slots (5 V); two 64-bit, 66-MHz PCI slots (3.3V); two 64-bit, 100-MHz PCI-X slots (3.3V)
- 450 W Dual Redundant Power Supply
- 1-512 MB DDR266, ECC SDRAM
- EIDE CDRW/DVD-ROM Drive
- Two 36 GB or larger, 15K RPM, Ultra 320 SCSI Hard Drives
- 3.5" 1.44 Mb floppy drive
- Integrated ATI Rage XL video controller; VGA connector - Video Memory: 8 MB
- PERC4-SC SCSI RAID-1 Controller
- PowerVault 100T, DDS4 Tape Drive

PE1600 options

The following is a list of the additional options that may be configured in your platform:

- LCNP4
- For Quad video – add a second dual video card (PCI)
- SCSI Controller
- Dual NIC for FTE
- Remote Power
- Two additional 512 MB memory modules

1.3 PE1800 Description

Enclosure

The PE1800 is the PC platform used for TPS and Experion PKS nodes, and can be mounted in cabinet furniture or used as a desktop.

When mounted in a cabinet, the enclosure uses the following amount of space:

- Fixed mount: 5U (8.75”) of rack space
- Slide mount: 6U (10.5”) of rack space

PE1800 options

The PE1800 is available in the two configurations, each of which will run the latest versions of the Application Processing Platform (APP Node) and Experion Server, ESVT, ESV, ACE, EHG.

- Experion server platform dual CPU: Intel Xeon 3.4 GHz x 2, 1MB L2 Cache with one 36GB 15K RPM hard disks and 1GB DDR2 400MHz ECC memory; non-RAID and a 650W non-redundant power supply- 120/240 VAC.
- Experion server platform dual CPU: Intel Xeon 3.4 GHz x 2, 1MB L2 Cache with two 36GB 15K RPM Hard Disks & 1GB DDR2 400MHz ECC memory, RAID 1 and dual swappable redundant 675 watt power supplies - 120/240 VAC.

Electronics module for PE1800

The peripheral electronics assemblies are based upon the Peripheral Component Interconnect (PCI-32/64), PCIX and PCIE bus, USB 1.1 or USB 2.0. The standard DDR2 memory for this platform is 1GB (2 – 512 MB DDR2 single ranked DIMMS), optionally expandable to 3.0 GB. Memory devices must be from the same memory supplier. The system has 1MB of L2 Cache memory.

Storage and media devices

All mass storage devices attach using SATA or SCSI connectors. The floppy drive attaches to the motherboard with a standard floppy disk connector and EIDE CDRW/DVD combination drive attaches to the motherboard with an IDE connector. The RAID-1 SCSI controller card in PCIX slot 5 has one SCSI connector for the hard drives. The standard mass storage for this platform is one 36 GB SCSI hard drive available in a RAID1 (mirrored) or non-RAID configuration.

PE1800 common standard features

The following is a list of the common features of this platform:

- Dell motherboard with Intel Xeon Dual CPU's
- Integrated Gigabit Ethernet NIC
- Integrated Single-Channel U320 SCSI Controller (supports up to six drives)
- Integrated SATA connectors (2) and SATA controller
- Expansion slots: Six total - Two PCI Express slots, two PCI-X slots (64-bit/100MHz), one PCI (32-bit/33MHz, 5v), and one PCI (64-bit/66MHz).
- 1MB L2 Cache
- Ports: 1 serial, 1 parallel, 4 USB V2.0, 2 in front and 2 in back
- 1GB DDR2 400MHz, ECC SDRAM
- EIDE CDRW/DVD-ROM Drive
- One or Two 36 GB or larger, 15K RPM, Ultra 320 SCSI Hard Drives
- .5" 1.44 Mb floppy drive
- Integrated ATI Radeon 7000-M video controller w/16MB of memory
- PERC4-SC SCSI RAID-1 Controller (for RAID 1 system)
- PowerVault 100T, TRAVAN Tape Drive
- Two hot-pluggable 675W power supplies in a 1 + 1 redundant configuration (RAID1) or one 650W Non-redundant power supply (Non-RAID). PE1800 options

The following is a list of the additional options that may be configured in your platform:

- LCNP4 (5-volt legacy)
- Dual NIC for FTE
- Single NIC
- ControlNet (five-volt legacy)
- Universal ControlNet (3.3-volt to 5-volt)
- LCNP4M (3.3-volt to 5-volt)
- SCSI RAID controller

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1.4. Finding Information for Your Platform

1.4 Finding Information for Your Platform

Honeywell documentation

The following table lists other Honeywell publications that may be useful when installing or operating the PE1600/PE1800 platform.

Table 1-1 Honeywell Publications

Publication	Contains information on
<i>FE05: Fault Tolerant Ethernet Installation and Service Guide</i>	Contains information for installing and using FTE on a TPS or Experion PKS node.

Dell PE1600 SC documentation

The following table lists Dell publications and other sources of information that will be useful when installing, operating and servicing the PE1600SC.

Table 1-2 Dell PE1600SC Publications

Publication	Contains information on	Available
<i>Information Update</i>	Last-minute updates about technical changes to your computer or advanced technical-reference material for experienced users or technicians.	Packaged with the computer
<i>Dell™ PowerEdge™ System Information Guide</i>	Warranty information Safety information	Packaged with the computer www.dell.com
<i>Setting Up Your System</i>	Unpacking and connecting cables	Packaged with the computer
<i>Quick Installation Guide</i>	Installing and configuring the server and operating system	Packaged with the computer Product Documentation CD
<i>Dell™ PowerEdge™ 1600SC Systems User's Guide</i>	How to remove and replace parts Technical specifications How to configure system settings	Product Documentation CD www.dell.com

1. System Planning

1.4. Finding Information for Your Platform

Publication	Contains information on	Available
	How to troubleshoot and solve problems	
<i>Dell™ PowerEdge™ Operating System Driver Installation Guide</i>	Installing system drivers	Packaged with the computer www.dell.com
<i>PERC 4/Di User's Guide</i>	Using and configuring the RAID Controller	Packaged with the computer www.dell.com
<i>Dell™ PowerEdge™ 1600SC Systems Installation and Troubleshooting Guide</i>	Diagnosing problems Using status indicators for troubleshooting	Product Documentation CD www.dell.com

Dell PE1800 documentation

The following table lists Dell publications and other sources of information that will be useful when installing, operating and servicing the PE1800.

Table 1-3 Dell PE1800 Publications

Publication	Contains information on	Available
<i>Information Update</i>	Last-minute updates about technical changes to your computer or advanced technical-reference material for experienced users or technicians.	Packaged with the computer
<i>Dell™ PowerEdge™ System Information Guide</i>	Warranty information Safety information	Packaged with the computer www.dell.com
<i>Setting Up Your System</i>	Unpacking and connecting cables	Packaged with the computer
<i>Quick Installation Guide</i>	Installing and configuring the server and operating system	Packaged with the computer Product Documentation CD
<i>Dell™ PowerEdge™ 1800 Systems User's Guide</i>	How to remove and replace parts Technical specifications How to configure system	Product Documentation CD www.dell.com

1. System Planning

1.4. Finding Information for Your Platform

Publication	Contains information on	Available
	settings How to troubleshoot and solve problems	
<i>Dell™ PowerEdge™ Operating System Driver Installation Guide</i>	Installing system drivers	Packaged with the computer www.dell.com
<i>Dell™ PowerEdge™ 1800 Systems Installation and Troubleshooting Guide</i>	Diagnosing problems Using status indicators for troubleshooting	Product Documentation CD www.dell.com

1.5 PE1600 Slot Requirements

Overview

In addition to the standard configuration for the PE1600, your platform may be configured with additional options based on the model number you ordered. This section contains slot and configuration requirements for additional options.

PE1600 slot requirements

The following table identifies the specific slot requirements for each of the Honeywell optional items.

Option	Slot Description
SCSI Controller	PCI1
Free slot	PCI2
Free slot	PCI3
Free slot	PCI4
LCNP4	PCI5
Dual NIC	PCI6

Memory upgrade configuration

The standard installed memory for the PE1600 is 512 MB, but it can be expanded up to 2 GB by installing additional 512 MB memory modules in the DIMM sockets as shown in the following table. Memory devices must be from the same supplier.

Table 1-4 Memory Upgrade Configuration

DIMM Socket	Memory Size	Total Memory
DIMMA	512 MB	512 MB
DIMMB	512 MB	1.0 GB
DIMMC	512 MB	1.5 GB
DIMMD	512 MB	2.0 GB

Desktop and cabinet options

The following table lists the desktop and cabinet model numbers for the PE1600.

Furniture Option	Model Number	Description
Base units	MZ-NTPC08	Desktop server (220V)
	MZ-NTPC08	Desktop server (120V)
Fixed rail cabinet options (includes mounting hardware)	TP-SPCMF1-101	In cabinet with KVM using fixed Rail Mounting Kit 120V AC
	TP-SPCMF1-201	In cabinet with Fixed Rail Mounting Kit 240V AC
	TP-SPCMF1-301	Short-haul remote with Fixed Rail Mounting Kit 120V AC
	TP-SPCMF1-401	Short-haul remote with Fixed Rail Mounting Kit 240V AC
Slide tray cabinet options (includes mounting hardware)	TP-SPCMS1-101	In-cabinet with KVM using Slide Tray Mounting Kit 120V AC
	TP-SPCMS1-201	In-cabinet with KVM using Slide Tray Mounting Kit 240V AC
	TP-SPCMS1-301	Short-haul remote using Slide Tray Mounting Kit 120V AC
	TP-SPCMS1-401	Short-haul remote using Slide Tray Mounting Kit 240V AC

1.6 PE1800 Slot Requirements

Overview

In addition to the standard configuration for the PE1800, your platform may be configured with additional options based on the model number you ordered. This section contains slot and configuration requirements for additional options.

Control Net configurations

The following tables identify the specific slot requirements for Control NET configurations.

Option 1 – Legacy 5 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	PCIX5
CNET (ABPCIC 01)	PCI4
Free Slot	PCIE3
Free Slot	PCIE2
Intel Pro/1000 M Server single port (optional)	PCI1

Option 2 – New 3.3 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M dual port, or Intel Pro/1000 M Server single port (optional)	PCIX5
Intel Pro/1000 M Server single port (optional)	PCI4
Free slot	PCIE3

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1.6. PE1800 Slot Requirements

Free slot	PCle2
CNET (ABPCIC 02	PCI1

LCNP4 configurations

The following tables identify the specific slot requirements for LCNP4 configurations.

Option 1 – Legacy 5 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	PCIX5
LCNP4	PCI4
Free slot	PCIE3
Free slot	PCIE2
Free slot	PCI1

Option 2 – New 3.3 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	PCIX5
LCNP4M	PCI4
Free slot	PCIE3
Free slot	PCIE2
Free slot	PCI1

FTE Control configurations

The following table identifies the specific slot requirement for an FTE configuration.

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port (optional)	PCIX5
Free slot	PCI4

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1.6. PE1800 Slot Requirements

Free slot	PCle3
Free slot	PCle2
Free slot	PCI1

SCADA configurations

The following tables identify the specific slot requirement for SCADA configurations.

Option 1 – Serial / Terminal Server Legacy 5 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	PCIX5
Stallion	PCI4
Free slot	PCle3
Free slot	PCle2
Intel Pro/1000 M dual port (optional)	PCI1

Option 2 – ModBus Legacy 5 Volt Card

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	PCIX5
Modicon card	PCI4
Free slot	PCle3
Free slot	PCle2
Intel Pro/1000 M dual port (optional)	PCI1

General Ethernet configurations

The following tables identifies specific slot requirement for general Ethernet configurations.

Option 1 – Single Ethernet (on-board controller is used)

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Free slot	PCIX5
	PCI4
Free slot	PCle3
Free slot	PCle2
Free slot	PCI1

Option 2 – DUAL Ethernet (on-board controller is used plus board listed)

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Server (single port)	PCIX5
Free slot	PCI4
Free slot	PCle3
Free slot	PCle2
Free slot	PCI1

Option 3 – Triple Ethernet (On-Board plus two single boards)

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M Server (single port)	PCIX5
Free slot	PCI4
Free slot	PCle3
Free slot	PCle2

1. System Planning
1.6. PE1800 Slot Requirements

Intel Pro/1000 M Server (single port)	PCI1
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Option 4 – Triple Ethernet (On-Board plus one dual port board)

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M dual port	PCIX5
Free slot	PCI4
Free slot	PCle3
Free slot	PCle2
Free slot	PCI1

Option 5 – Quad Ethernet (On-Board plus boards listed)

Option	Slot Description
PERC 4/SC (optional)	PCIX6
Intel Pro/1000 M dual port	PCIX5
Free slot	PCI4
Free slot	PCle3
Free slot	PCle2
Intel Pro/1000 M Server single port)	PCI1

Memory upgrade configuration

The standard installed memory for the PE1800 is 1 GB (two 512 MB DDR2 400 MHz SDRAM), but it can be expanded up to 3 GB by installing additional 512 MB memory modules in the DIMM sockets as shown in the following tables. Memory devices must be from the same supplier.

Table 1-5 2 GB Memory Upgrade Configuration

DIMM Socket	Memory Size	Total Memory
DIMM1B	512 MB	512 MB
DIMM1A	512 MB	1.0 GB
DIMM2B	512 MB	1.5 GB
DIMM2A	512 MB	2.0 GB
DIMM3B	Disabled*	
DIMM3A	Disabled*	

*Limitations within the motherboard's chipset do not extend usage to these memory sockets.

Table 1-6 3 GB Memory Upgrade Configuration

DIMM Socket	Memory Size	Total Memory
DIMM1B	512 MB	512 MB
DIMM1A	512 MB	1.0 GB
DIMM2B	1 GB	2 GB
DIMM2A	1 GB	3 GB
DIMM3B	Disabled*	
DIMM3A	Disabled*	

*Limitations within the motherboard's chipset do not extend usage to these memory sockets.

1.7 Industrial Regulatory Compliance

Overview

The compliance specifications in this section apply to TPS and Experion PKS Cabinets.



WARNING

Honeywell does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

Electromagnetic Compatibility (EMC)

Table 1-7 EMC Specifications (Industrial Regulatory)

European Community:

Emissions: IEC 61326, 1997 (Industrial Locations, CISPR 11, Class A)

Immunity: IEC 61326, 1997 (Industrial Locations)

Attention: The following formula is a proximity guideline, for use of Portable Transceivers

(walkie-talkies), in the frequency range of 80MHz to 1GHz:

$D > 0.30 \cdot \sqrt{P}$ (D must be greater than 0.30 multiplied by the square root of P)

D = Distance from equipment, in meters.

P = Power Output of the Portable Transceivers (walkie-talkies), in Watts.

Examples:

P = 10 Watts, D > 0.949 meters

P = 5 Watts, D > 0.671 meters

P = 1 Watt, D > 0.300meters

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables.

(360 degree metallic shielding), in order to comply with the above EMC standards.

Safety compliance

Table 1-8 Safety Compliance (Industrial Regulatory)

Product Safety Compliance:

CSA C22.2 No. 1010.1-92(R1999) & 1010.1B-97 (R2001) Am. 2

IEC 61010-1, 2001, 2nd edition

Note: Within the above referenced standards is a “Normative Reference” section citing additional standards, which may apply as, suited and required for product compliance.

1.8 Light Industrial Regulatory Compliance

Overview

The compliance specifications in this section apply to the following platforms:

- TPS and Experion
- Desktop
- Desk-side
- Classic Console
- Z-Console
- ICON Console Systems



WARNING

Honeywell does not claim Safety Compliance or Electromagnetic Compatibility (EMC) Compliance for system equipment configurations that have not been described in this manual as standard system configurations. Any equipment configuration other than that described in this publication decertifies the Safety and EMC compliance of this product.

1. System Planning

1.8. Light Industrial Regulatory Compliance

Electromagnetic Compatibility (EMC)

Table 1-9 EMC Specifications (Light Industrial Regulatory)

European Community:

Emissions: IEC 61326, 1997 (Basic Requirements, CISPR11, Class A)

Immunity: IEC 61326, 1997 (Basic Requirements)

Attention: The following formula is a proximity guideline, for use of Portable Transceivers

(walkie-talkies), in the frequency range of 80MHz to 1GHz:

$D > \sqrt{P}$ (D must be greater than the square root of P)

D = Distance from equipment, in meters.

P = Power Output of the Portable Transceivers (walkie-talkies), in Watts.

Examples:

P = 10 Watts, D > 3.162 meters

P = 5 Watts, D > 2.236 meters

P = 1 Watt, D > 1.000 meters

Note: Electrical cables, which are routed external to the equipment, must be fully shielded cables (360 degree metallic shielding), in order to comply with the above EMC standards.

Safety Compliance

Table 1-10 Safety Compliance (Light Industrial Regulatory)

Product Safety Compliance:

CSA C22.2 No. 1010.1-92(R1999) & 1010.1B-97 (R2001) Am. 2

IEC 61010-1, 2001, 2nd edition

Note: Within the above referenced standards is a "Normative Reference" section citing additional standards, which may apply as, suited and required for product compliance.

1. System Planning

1.8. Light Industrial Regulatory Compliance

2. Platform Installation

2.1 Introduction

Overview

This section contains procedures for installing and cabling the PE1600/PE1800 as a desktop unit or in the Honeywell cabinet.

Tasks for installing the PE1600/PE1800

The following table lists the major platform installation tasks.

Table 2-1 PE1600 Installation Tasks

Task	For more information see . . .
✓ Be aware of all power and grounding requirements for your furniture.	Specific site requirements 2.2 "Power and Grounding Requirements" <i>TPN System Installation (SW20-600)</i> , Section 4, "System Grounding"
✓ Install the platform in the cabinet.	2.4 "Installing the PE1600/PE1800 in a MarkHon Cabinet" 2.5 "Using Adapters with the Platform"
✓ Start up the PE1600	2.6 "Starting up the Platform"
✓ If you are remoting the computer, connect the remote components.	<i>Remote Peripheral Systems Installation and Upgrade (RE01)</i>

Before you begin

Before performing the procedures in this section, perform or verify the following tasks have been performed.

✓	Description
	Verify the cabinet has been properly grounded.
	Unpack the platform from the box and verify all parts are accounted for.
	Have available a Philips head screwdriver.
	Position the furniture in the area it will be used.
	Identify and verify all necessary cables for your particular configuration are available.

2. Platform Installation

2.2. Power and Grounding Requirements

2.2 Power and Grounding Requirements

Grounding for PC based nodes

The ground connection is made through the third wire in the AC power cord.

Grounding consoles and cabinets

The *TPN System Installation* manual, Section 4, “System Grounding” contains information on grounding furniture, including the following:

- Ground Wiring Overview
- Grounding LCN Cabinets and Stations
- Cabinet Logic Ground
- Grounding LCN Cables
- Grounding Stations in Z-Console Furniture

AC Power Warning



53 89 5



WARNING

The power supply circuit is connected to AC power when the power cable is connected. The power control switch on the front panel only enables the power supply circuit outputs.



ATTENTION

It is strongly recommended that the power cord be connected to a clean power source with backup such as an Uninterruptible Power Source (UPS)..

2.3 Connecting Platform Cables

PE1600 standard connections

The following picture shows the back of the PE1600 enclosure with standard options.

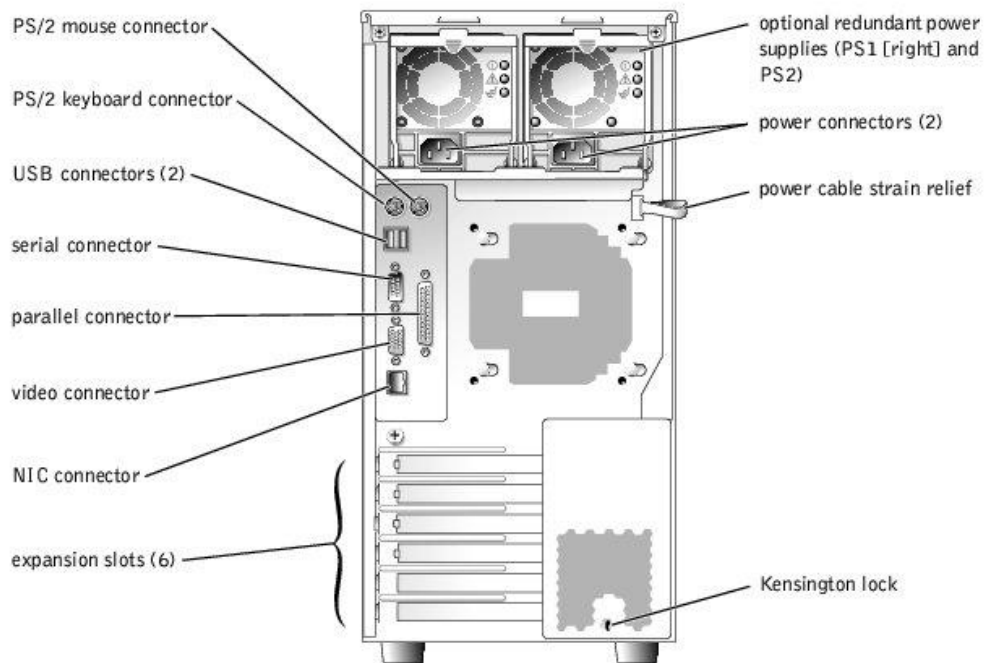


Figure 2-1 PE1600 Rear View

2. Platform Installation

2.3. Connecting Platform Cables

PE1600 expansion slot configuration

The following table describes the configuration requirements for the PE1600 expansion slots.

Table 2-2 PE1600 Expansion Slot Configuration

Slot	Option	Location (vertical)	Location (horizontal)
PCI1	SCSI Controller	Top	Far left
PCI2	Free slot	2nd from top	2nd from left
PCI3	Free slot	3rd from top	3rd from left
PCI4	Free slot	3rd from bottom	3rd from right
PCI5	LCNP4	2nd from bottom	2nd from right
PCI6	Dual NIC	Bottom	Far right

PE1800 standard connections

The following picture shows the back of the PE1800 enclosure with standard options.

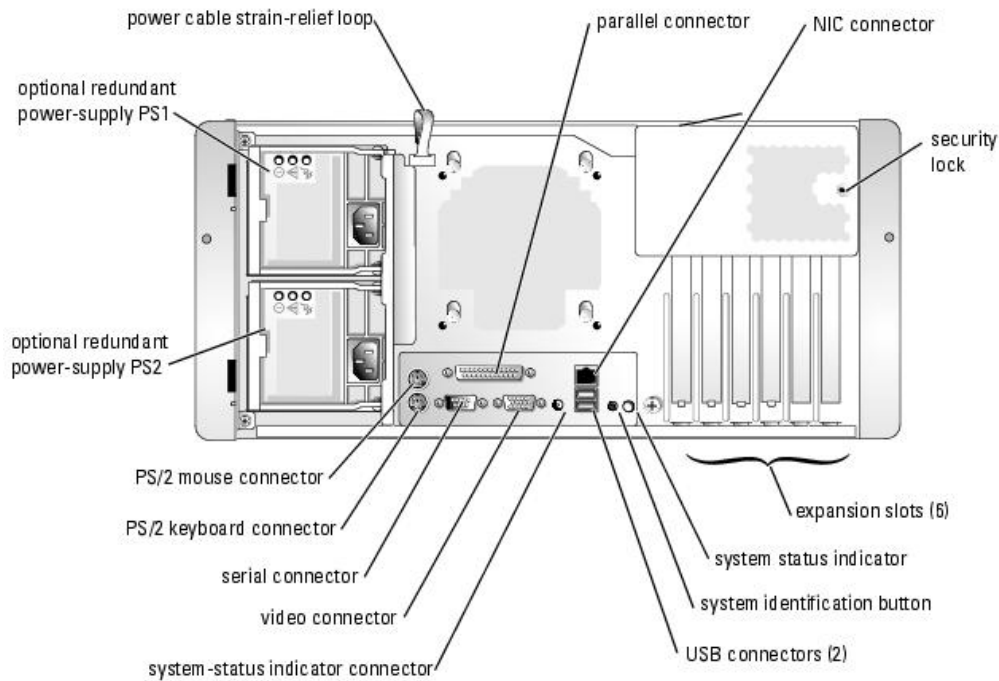


Figure 2-2 PE1800 Rear View

PE1800 expansion slot Control Net configurations

The following table describes the Control Net configuration requirements for the PE1800 expansion slots.

Table 2-3 PE1800 Control Net Slot Configuration

Slot	New 3.3 volt Option	Legacy 5-volt Option	Location (vertical)	Location (horizontal)
PCI1	Intel Pro/1000 M Server single port (optional)	Intel Pro/1000 M Server single port (optional)	Top	Far left
PCI2	Free Slot	Free Slot	2nd from top	2nd from left
PCI3	Free Slot	Free Slot	3rd from top	3rd from left
PCI4	CNET (ABPCIC 01)	CNET (ABPCIC 01)	3rd from bottom	3rd from right
PCIX-5	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	PERC 4/SC (optional)	Bottom	Far right

2. Platform Installation

2.3. Connecting Platform Cables

PE1800 expansion slot LCNP4 configurations

The following table describes the LCNP4 configuration requirements for the PE1800 expansion slots.

Table 2-4 PE1800 LCNP4 Slot Configuration

Slot	New 3.3 volt Option	Legacy 5-volt Option	Location (vertical)	Location (horizontal)
PCI1	Free Slot	Intel Pro/1000 M Server single port (optional)	Top	Far left
PCI2	Free Slot	Free Slot	2nd from top	2nd from left
PCI3	Free Slot	Free Slot	3rd from top	3rd from left
PCI4	LCNP4M	LCNP4	3rd from bottom	3rd from right
PCIX-5	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	PERC 4/SC (optional)	Bottom	Far right

PE1800 expansion slot FTE configurations

The following table describes the FTE configuration requirements for the PE1800 expansion slots.

Table 2-5 PE1800 FTE Slot Configuration

Slot	New 3.3 volt Option	Location (vertical)	Location (horizontal)
PCI1	Free Slot	Top	Far left
PCI2	Free Slot	2nd from top	2nd from left
PCI3	Free Slot	3rd from top	3rd from left
PCI4	Free Slot	3rd from bottom	3rd from right
PCIX-5	Intel Pro/1000 M Dual Port (optional)	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	Bottom	Far right

PE1800 expansion slot SCADA configurations

The following table describes the SCADA configuration requirements for the PE1800 expansion slots.

Table 2-6 PE1800 SCADA Slot Configuration

Slot	Serial/terminal server legacy 5-volt card	ModBus legacy 5-volt card	Location (vertical)	Location (horizontal)
PCI1	Intel Pro/1000 M dual port (optional)	Intel Pro/1000 M dual port (optional)	Top	Far left
PCI2	Free Slot	Free Slot	2nd from top	2nd from left
PCI3	Free Slot	Free Slot	3rd from top	3rd from left
PCI4	Stallion	Modicon card	3rd from bottom	3rd from right
PCIX-5	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	Intel Pro/1000 M Dual Port, or Intel Pro/1000 M Server single port (optional)	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	PERC 4/SC (optional)	Bottom	Far right

PE1800 expansion slot general Ethernet configurations

The following tables describe the general Ethernet configuration requirements for the PE1800 expansion slots.

Table 2-7 PE1800 Single/Dual Ethernet Slot Configuration

Slot	Single Ethernet	Dual Ethernet	Location (vertical)	Location (horizontal)
PCI1	Free slot	Free slot	Top	Far left
PCI2	Free slot	Free slot	2nd from top	2nd from left
PCI3	Free slot	Free slot	3rd from top	3rd from left
PCI4	Free slot	Free slot	3rd from bottom	3rd from right
PCIX-5	Free slot	Intel Pro/1000 M Server (single port)	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	PERC 4/SC (optional)	Bottom	Far right

2. Platform Installation

2.3. Connecting Platform Cables

Table 2-8 PE1800 Triple Ethernet Slot Configuration

Slot	Triple Ethernet	Triple Ethernet	Quad Ethernet	Location (vertical)	Location (horizontal)
PCI1	Intel Pro/1000 M Server (single port)	Free slot	Intel Pro/1000 M Server (single port)	Top	Far left
PCI2	Free slot	Free slot		2nd from top	2nd from left
PCI3	Free slot	Free slot		3rd from top	3rd from left
PCI4	Free slot	Free slot		3rd from bottom	3rd from right
PCIX-5	Intel Pro/1000 M Server (single port)	Intel Pro/1000 M dual port	Intel Pro/1000 M dual port	2nd from bottom	2nd from right
PCIX-6	PERC 4/SC (optional)	PERC 4/SC (optional)		Bottom	Far right

Power cables

The following table lists the AC power cables for the different PE1600 configurations

	Model Number	Description	Part Description(s)	Part Number(s)
Base units	MZ-NTPC08	Desktop server (220V)	50 HZ/240V power cord	(2) 51305557-100
	MZ-NTPC08	Desktop server (120V)	60 HZ/120V power cord	(2) 51107941-113
Fixed rail cabinet options (includes mounting hardware)	TP-SPCMF1-101	In cabinet with KVM using fixed Rail Mounting Kit 120V AC	120V AC power cord	(2) 51305490-400
	TP-SPCMF1-201	In cabinet with Fixed Rail Mounting Kit 240V AC	240V AC power cord	(2) 51305489-400
	TP-SPCMF1-301	Short-haul remote with Fixed Rail Mounting Kit 120V AC	120V AC power cord	(2) 51305490-400
	TP-SPCMF1-401	Short-haul remote with Fixed Rail Mounting Kit 240V AC	240V AC power cord	(2) 51305489-400
Slide tray cabinet options (includes mounting hardware)	TP-SPCMS1-101	In-cabinet with KVM using Slide Tray Mounting Kit 120V AC	120V AC power cord	(2) 51305490-500
	TP-SPCMS1-201	In-cabinet with KVM using Slide Tray Mounting Kit 240V AC	240V AC power cord	(2) 51305489-500
	TP-SPCMS1-301	Short-haul remote using Slide Tray Mounting Kit 120V AC	120V AC power cord	(2) 51305490-500
	TP-SPCMS1-401	Short-haul remote using Slide Tray Mounting Kit 240V AC	240V AC power cord	(2) 51305489-500

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

2.4 Installing the PE1600/PE1800 in a MarkHon Cabinet

Overview

The platform is mounted in a cabinet in a horizontal position. The human interface for cabinet mounted computers is either accessed locally in the cabinet or remotely using one of Honeywell's remoting options.

Mounting devices in cabinet

Servers can be mounted in a cabinet using the fixed or slide mount options. Due to safety and center-of-gravity considerations, servers must be mounted at the bottom of the cabinet, with a maximum of two servers in a cabinet. The various devices mounted in the cabinet require RETMA standard cabinet openings as specified in the following table.

Option	Rack Space Needed	Mounting Requirements
Slide mounted PE1600	6U 10.5 inches 26.7 cm	Any unused space must have a blank front panel and air duct baffle installed in the unused space to ensure proper airflow in the cabinet. Due to thermal constraints, only 6 total computers per cabinet are allowed
Fixed mounted PE1600	5U 8.75 inches 22.2cm	
Human interface: 4 port KVM switch; 17-inch CRT; keyboard & cursor device	12 U 21 inches 55.4 cm	Mount in rack space 13U through 24U to ensure the keyboard is at the proper height
Human interface: 8 port KVM switch; 15-inch LCD; keyboard & cursor device	1U 1.88 inches 4.78 cm	Mount in rack space 13U on telescopic slides
MAU	None	Mount to the back side of the panduit cable trough using the MAU mounting bracket.

Unused cabinet spaces



ATTENTION

To ensure proper computer cooling and airflow through the cabinet, all unused rack mount locations must have a blank front panel and air duct baffle installed.

Example rack mounting locations using slide mounts

The following table is an example of a cabinet configuration using the slide mount option. The **Rack Mount** column lists the RETMA standard units in the cabinet.

Component Description	Rack Mount
	40U
5U high slide mount for Dell WS360/370 workstation (short-haul remote)	39U
	38U
	37U
	36U
	35U
5U high slide mount kit for Dell WS360/370 workstation (with short-haul remote option)	34U
	33U
	32U
	31U
	30U
5U high slide mount for Dell WS360/370 workstation (with short-haul remote option)	29U
	28U
	27U
	26U
	25U
12U enclosure for 17-inch monitor, keyboard, mouse and 4-port KVM switch	24U
	23U
	22U
	21U
	20U
	19U
	18U
	17U
	16U
	15U
6U high slide mount for Dell PE1600SC/Dell PE1800 server (KVM human interface option)	14U
	13U
	12U
	11U
	10U
6U high slide mount for Dell PE1600SC/PE1800 server (KVM human interface option)	9U
	8U
	7U
	6U
	5U
	4U
	3U
	2U
	1U

Note: Short haul remote electronic box is mounted on brackets that attach to the rear of the computer.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Example rack mounting locations using fixed mounts

Using all fixed mounts, up to two Dell PE1600SC servers may be mounted in the first 10U of rack mounting space, starting at the bottom of the cabinet. The following table is an example of a cabinet configuration using the fixed mount option. The **Rack Mount** column lists the RETMA standard units in the cabinet.

Component Description	Rack mount
4U high fixed mount for Dell WS360/370 workstation (with short haul remote option)	40U
	39u
	38u
	37U
4U high fixed mount for Dell WS360/370 workstation (with short haul remote option)	36U
	35U
	34U
	33U
4U high fixed mount for Dell WS360/370 workstation (with short haul remote option)	32U
	31U
	30U
	29U
4U high fixed mount for Dell WS360/370 workstation (with short haul remote option)	28U
	27U
	26U
	25U
12U enclosure for 17-inch monitor, keyboard, mouse and 4-port KVM switch	24U
	23U
	22U
	21U
	20U
	19U
	18U
	17U
	16U
	15U
	14U
	13U
	12U
	11U
5U high fixed mount for Dell PE1600 SC server (KVM human interface)	10U
	9U
	8U
	7U
5U high fixed mount for Dell PE1600 SC server (KVM human interface)	6U
	5U
	4U
	3U
	2U
	1U

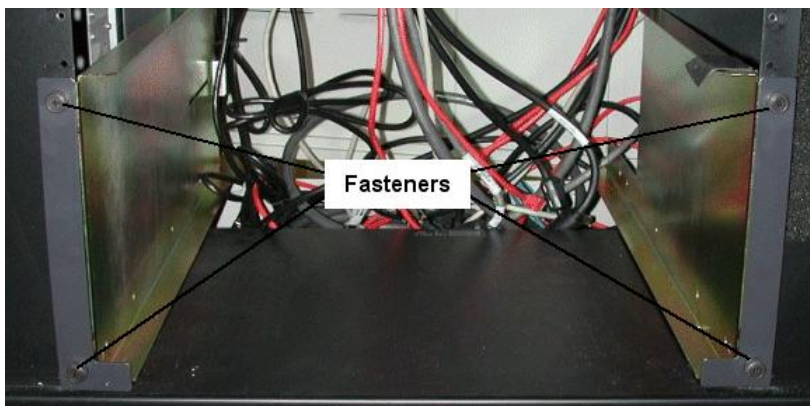
Note: Short haul remote electronic box is mounted on brackets that attach to the rear of the computer.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Install computer in cabinet using the fixed mount option

Use the following procedure to install the PE1600 platform in the cabinet using the fixed mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them. 
3	Verify that the cabinet mounting rails are clear of any installed cables.
4	Using two people, slide the computer all the way into the cabinet enclosure between the mounting railings.
5	Replace the two mounting bracket covers and tighten the four screws.
	ATTENTION The following steps cover the mounting of Honeywell components in the cabinet some of which may not apply to your particular configuration.
6	If you are using Honeywell's remote option, from the rear of the cabinet, install the device according to the instructions packaged with the device.
7	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

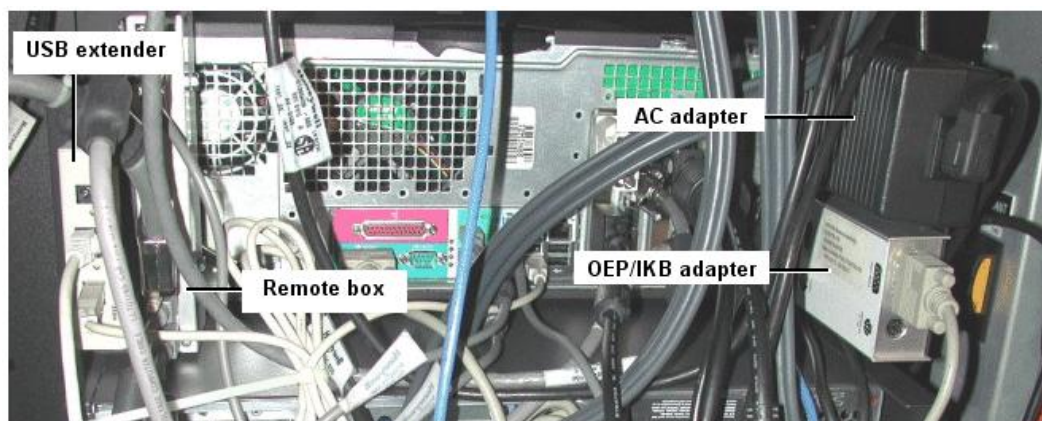


Figure 2-3 **Components Mounted in Cabinet**

Install computer in cabinet using the slide mount option

Use the following procedure to install the platform in the cabinet using the slide mount option.

Step	Action
1	Open the cabinet door to access the inside of cabinet.
2	If the retaining brackets are attached, loosen the two captivated fasteners on each bracket and remove them.
3	Verify that the slide mount bottom and sides are clear of any installed cables.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Step	Action
4	Using the mounting tray handle, pull the tray to its fully extended and locked position.  A photograph showing a person's hands pulling a black mounting tray out of a server cabinet. The tray is partially extended, and the person is using a handle on the front edge to pull it further out. The cabinet is open, revealing internal components and wiring. The tray is mounted on a metal rail system.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

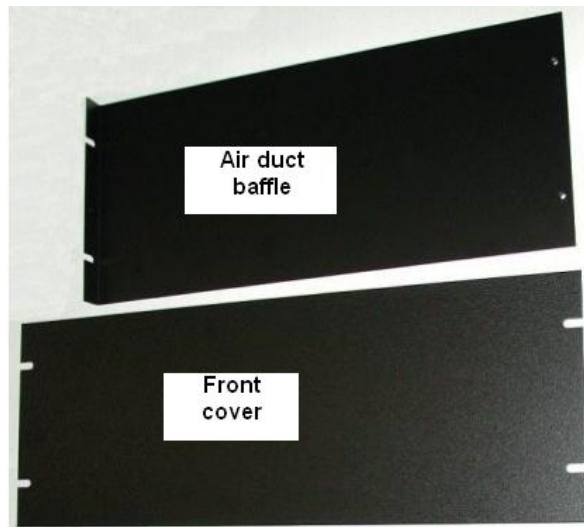
Step	Action
5	Using two people, place the computer on the mounting tray so the bottom surface of the front bezel is aligned with the front surface of the slide tray and slide the tray back into the console. 
6	Replace the two retaining brackets and tighten the four captivated fasteners.
7	From the rear of the cabinet, slide the PC retainer bracket (L-shaped angle bracket) forward in its slot until it contacts the back surface of the PC chassis. Tighten the lock nut.
	ATTENTION The following steps cover the mounting of Honeywell components in the cabinet some of which may not apply to your particular configuration.
8	If you are installing Honeywell's remote option, install it according to the instructions that came with the device.
9	Go to the "Install air duct baffles and blank panel front covers in cabinet" procedure in this section.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Install air duct baffles and blank panel front covers in cabinet

Use the following procedure to install the air duct baffle and blank panel front cover in the cabinet. Installing these parts will insure that the airflow within the cabinet allows proper cooling of the computers.



2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Step	Action
1	Place the air duct baffle inside the cabinet with the bent tab resting along the front of the right cabinet rail as shown in the picture. 
2	Attach the air duct baffle to the rear cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the cabinet rail. 
3	Place the blank panel across the front of the cabinet rails and attach it to the air duct baffle and right cabinet rail using two hex head screws and two washers; make sure the flat side of the washer touches the screw head and the washer teeth touch the panel.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Step	Action
	
4	Attach the other side of the blank panel to the left cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 
5	Place the 1 blank panel across the front of the cabinet rails and attach it to the air duct baffle and right cabinet rail using two hex head screws and two washers; make sure the flat side of the washer touches the screw head and the washer teeth touch the panel.

2. Platform Installation

2.4. Installing the PE1600/PE1800 in a MarkHon Cabinet

Step	Action
	
6	Attach the other side of the blank panel to the left cabinet rail using two hex head screws and two washers, making sure the flat side of the washer touches the screw head and the washer teeth touch the panel. 

Connect cables in the cabinet

Use this procedure to connect all cables to the computer platform. Refer also to Figure 2-1, PE1600 Rear View. If you are connecting the computer to a remote system, refer to the remote installation manual packaged with your remote system.

Step	Action
1	From the rear of the cabinet, open the door to access the computer.
2	Connect the mouse and keyboard cables to the PS/2 connectors.
3	If you are not using Fault Tolerant Ethernet (FTE), connect the Ethernet cable to the RJ-45 connector on the Network Interface Card. If you are using FTE, you will need to connect the FTE cables according to the instructions in the <i>FTE Installation and Service Guide</i> .
4	Connect the video cable.
5	Connect the LCN MAU cable to the MAU connector on the LCNP4 card, if one is present.
6	Secure any loose cables, and verify that all cables have proper strain relief.
7	Go to Section 2.6, "Starting up the Platform" to complete the installation.

2. Platform Installation

2.5. Using Adapters with the Platform

2.5 Using Adapters with the Platform

About the PE1600 serial ports

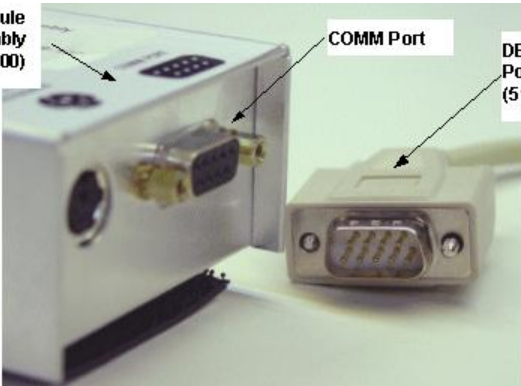
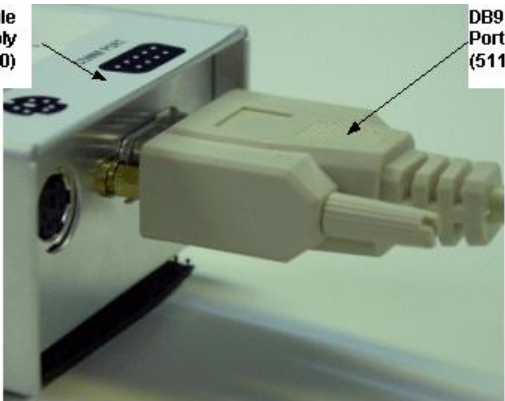
The PE1600 has two standard serial ports: COM 1 and COM 2. To ensure these ports work correctly with Honeywell hardware, the serial ports are reconfigured as COM 3 and COM 4 respectively when the computer is initialized. It is important that you connect Honeywell devices to the correct serial port as defined in the instructions. The following table summarizes the port identification and device requirements.

Table 2-9 COM Port Connections

Honeywell Device	Connect to this Serial Port	Example
OEP	COM1 (configured as COM3)	
IKB	COM1 (configured as COM3)	

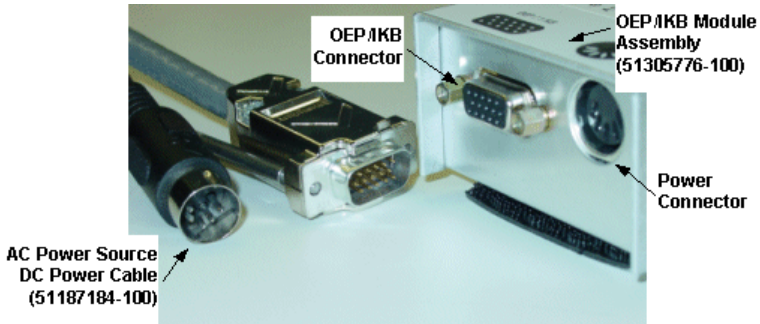
Connect the OEP/IKB adapter module to the device

Use this procedure to connect an IKB or OEP to the PE1600 platform. An OEP or IKB are always connected to the COM 1 port on the PE1600 platform.

Step	Action
1	Connect the male end of the DB9 M/F Comm Port Cable (51196990-200) to the COMM PORT on the OEP/IKB Module Assembly (51305776-100) as shown.  The diagram shows a white OEP/IKB Module Assembly with a DB9 connector labeled 'COMM Port'. A white DB9 M/F Comm Port Cable is being inserted into the port. Labels with arrows point to the 'OEP/IKB Module Assembly (51305776-100)', the 'COMM Port', and the 'DB9 M/F COMM Port Cable (51196990-200)'.
2	Tighten the thumbscrews on the COMM Port cable to secure the cable to the module assembly.  The diagram shows the DB9 M/F Comm Port Cable fully inserted into the COMM Port of the OEP/IKB Module Assembly. Two thumbscrews are shown being tightened to secure the cable. Labels with arrows point to the 'OEP/IKB Module Assembly (51305776-100)' and the 'DB9 M/F COMM Port Cable (51196990-200)'.

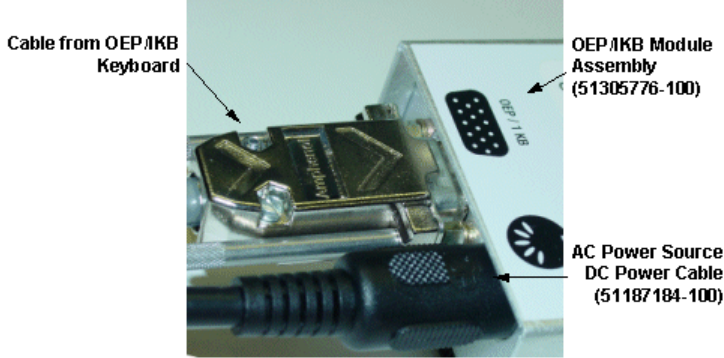
2. Platform Installation

2.5. Using Adapters with the Platform

Step	Action
3	If you are connecting an IKB, insert the round DIN plug of the IKB12 PS/2 cable (51305381) into the PS/2 Connector on the OEP/IKB Module Assembly (51305776-100).  OEP/IKB Module Assembly (51305776-100) IKB12 PS/2 KB Cable (51305381-300) DB9 M/F COMM Port Cable (51196990-200)
4	Insert the round DIN plug from the AC Power Source DC Power Cable (51187184-100) into the Power Connector on the OEP/IKB Module Assembly (51305776-100).  OEP/IKB Connector OEP/IKB Module Assembly (51305776-100) Power Connector AC Power Source DC Power Cable (51187184-100)

2. Platform Installation

2.5. Using Adapters with the Platform

Step	Action
5	Connect the proper end of the cable from the OEP/IKB Keyboard to the OEP/IKB Connector on the OEP/IKB Module Assembly (51305776-100) and tighten the thumbscrews on the OEP Keyboard cable to secure the cable to the module assembly. 
6	Connect the female end of the DB9 Comm Port Cable to the COM 1 port on the PE1600 serial ports.
7	Tighten the thumbscrews on the COMM Port cable to secure the cable to the computer.
8	If you are connecting an IKB, connect the other end of the IKB12 PS/2 cable (51305381-100) to the Keyboard PS/2 Port on the computer.
9	Connect the AC power cord (51305567) to the AC power source (51187184-100).
10	Plug in the AC power cord.

2.6 Starting up the Platform

Overview

The following table lists those tasks that must be performed prior to operating your platform. This section contains step-by-step procedures for each of these tasks.

✓	Task
	Turn on power and check status
	Check LCNP status
	Set monitor resolution

Turn on power and check status

Step	Action
1	Press the power button the front panel.
2	Wait for the power light to become solid green.
3	If the power light does not become solid green, refer to the <i>Installation and Troubleshooting Guide</i> .

Check LCNP status

If this is a TPS node with the LCNP4 board installed, verify that the LCNP passed self-test.

Step	Action
1	From the Start menu, select LCNP Status .
2	Verify that the LCNP status indicates Passed Self Test and the circle is green.
3	Verify that LCN Address appears in the LEDs field of the LCNP Status display. In the above example, 39 was entered as the LCN Address:



REFERENCE

Refer to the LCNP Status section in the *LCNP Status User's Guide* for more information.

Set monitor resolution

When the platform initialized the monitor configuration is established based on the following user input:

- CRT type monitors: 75 Hz
- FPD type monitors: 60 Hz

If, after initialization, you install a monitor type different than what was originally defined, you will need to adjust the monitor settings for optimal performance. See also your specific monitor user guide for other recommended settings.

Step	Action
1	Select Start > Settings > Control Panel .
2	Double-click the Display icon and select the Settings tab.
3	Under Screen area, drag the slider to the appropriate resolution.
4	Click Apply .

2. Platform Installation

2.6. Starting up the Platform

3. Platform Operation

3.1 Overview

Front view of PE1600SC enclosure

The following figure shows the front view of the PE1600SC enclosure. See also the publications listed in Table 1-2, Dell PE1600SC Publications.

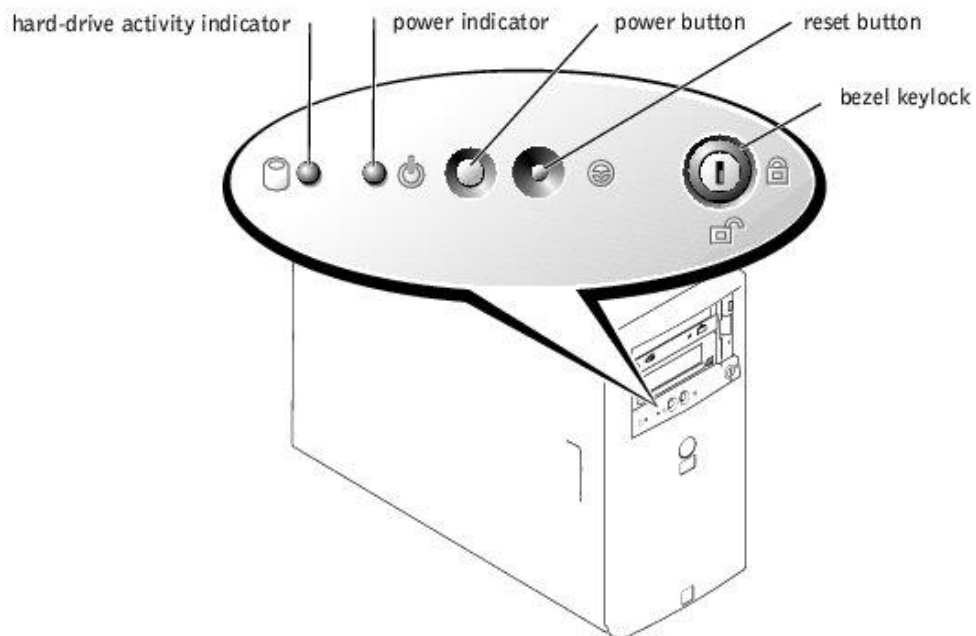


Figure 3-1 Front View of PE1600SC Enclosure

3. Platform Operation

3.1. Overview

Front view of PE1800 enclosure

The following figure shows the front view of the PE1800 enclosure. See also the publications listed in Table 1-3, Dell PE1800 Publications.

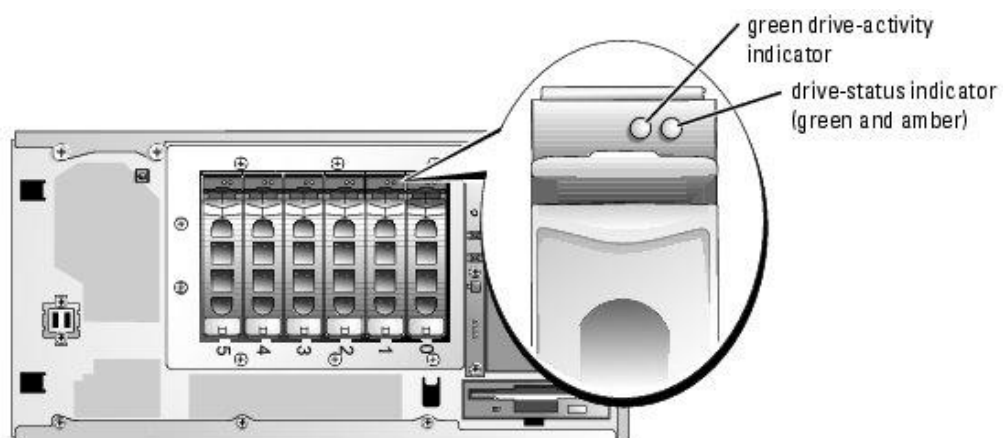


Figure 3-2 Front View of PE1800 Enclosure

3.2 Network Connections

Overview

Each platform must be connected to an LCN network and/or an ETHERNET network.

Ethernet network

ETHERNET 10/100/1000 Base T connection is standard on the Dell PE1600 platform. A Dual NIC option is available for FTE. If you are using FTE, you must install the FTE cables according to the specific instructions in the *FTE Installation and Service Guide*.

LCN network

The connection to the LCN is made via an optional Local Control Network Processor (LCNP4) card. This card provides the communication path for the Dell PE1600 to other LCN modules. The LCNP4 consists of an LCNP4 Card, a MAU cable and the LCN MAU (Media Access Unit). The LCNP4 card is a full length PCI card that consumes one PCI slot. The LCNP4, MAU and MAU cable are required to connect to the LCN Network.

The LCN node address should be set to the address the customer requires. If the LCN address is not known then the node address should be set to zero (0). Setting the address to zero (0) allows the node to be connected to the LCN without the risk of an address conflict with some other node. This is consistent with the current LCN standard procedure.

LCN cables

The two cables and T-connectors (and terminators, if applicable) must be located underneath (the future location) of the GUS unit, with an additional loop, 1 meter in length (with a T-connector), to be routed into the rear of the platform where the connection is made to the LCN MAU. The cable between the LCNP4 board and the LCN MAU is 2 meters in length. See Figure 3-3.

Refer to *LCN System Installation*, Subsection 3.6 for the rules and techniques of installing an LCN cable system.

3. Platform Operation

3.2. Network Connections

LCN connections

The LCN Cable A and Cable B connections are made through a single cable from the LCNP4 board to the LCN Media Access Unit (MAU) contained in a metal

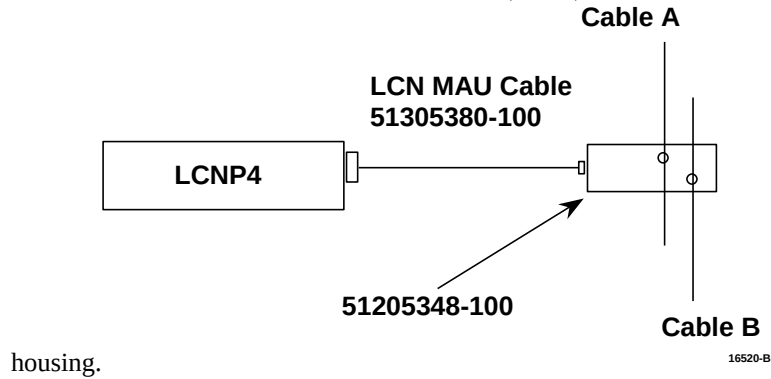


Figure 3-3 LCN MAU to LCN Cabling

MAU connection

Connect the MAU to both Cable A and Cable B coax T-connector as shown below.

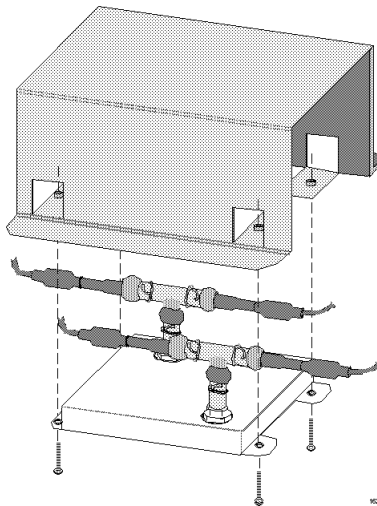


Figure 3-4 LCN MAU to LCN Cable T-Connections

4. Platform Servicing

4.1 Servicing the PE1600/PE1800 Enclosure

Before you begin servicing



Attention

Perform a complete system shutdown before you begin any of the procedures in this section.



CAUTION

Before you begin any of the procedures in this section, follow the safety instructions in the *Dell System Information*.



CAUTION

To avoid electrical shock, always unplug your computer from the electrical outlet before opening the cover.



Attention

Be careful when opening the computer cover to ensure that you do not accidentally disconnect cables from the system board.

Accessing the electronics

To simplify servicing, the PE1600 enclosure, when using the slide mount option, provides access to all internal hardware components without having to remove the enclosure from the Console or Cabinet in which it is mounted. See also your platform service manual.

Step	Action
1	If this is a slide mount computer, slide the tray from the furniture.
2	Use the system key to unlock the front bezel by placing the keylock in the open position.
3	Slide the latch in the direction of the arrow and pull the upper edge of the bezel away from the front panel.

4. Platform Servicing

4.1. Servicing the PE1600/PE1800 Enclosure

Step	Action
4	Pivot the bezel downward until it is at right angles to the front panel.
5	Unsnap the bezel from the metal clips on the front panel.
6	Loosen the three thumbscrews on the left-side cover.
7	Grasp the cover at both ends and slide it toward the front of the system.
8	Lift the cover away from the system.
9	If the computer is not already on its side, place it on its right side before servicing the components.
10	Service the hardware components as required: • For servicing the LCNP4 board, see Section 4.2 in this document. • For servicing other Honeywell installed options see Section 4.3 in this document. • For other components, see the platform service manual.

4.2 Servicing the LCNP4

LCNP4 board description

The LCNP4 board has 16 MB of on-board memory that is soldered to the board.

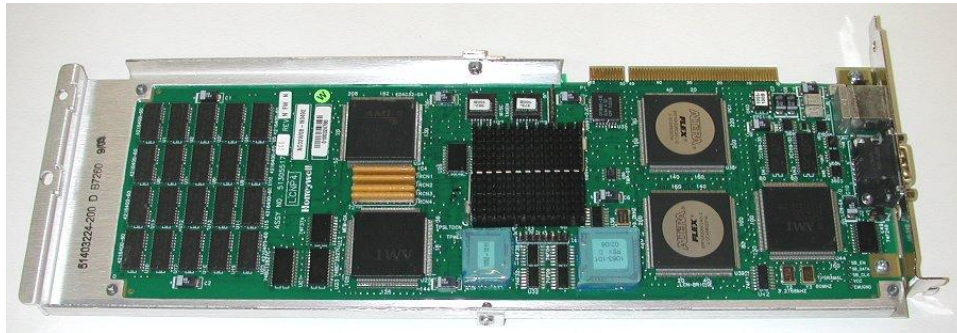


Figure 4-1 LCNP4 Board

Replace the LCNP4 board

Use the following procedure to replace the LCNP4 assembly. See also the *Dell™ PowerEdge™ 1600SC Systems Service Manual* or the *Dell™ PowerEdge™ 1800 Systems Service-Only Parts Replacement Procedures*.



ESD HAZARD

The LCNP4 board is an electrostatically sensitive device. Use a grounding strap and grounded work surfaces and equipment when handling this component. Store and transport parts only in electrostatically safe containers.

Step	Action
1	Perform all necessary procedures in Section 4.1, "Servicing the PE1600/PE1800 Enclosure" to access the LCNP4 assembly., including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure Disconnecting cables from the card being replaced.

4. Platform Servicing

4.2. Servicing the LCNP4

Step	Action
2	Turn off the system, including any attached peripherals, and disconnect the system from the electrical outlet.
3	Remove the bezel and the cover using the "Accessing the electronics" procedure in Section 4.1.
4	Disconnect the LCN MAU cable from the LCNP4 board.
5	Remove the filler bracket from the expansion slot.
6	While wearing a grounded ESD wrist strap, remove the LCNP4 board assembly from the PCI slot.
7	Verify that the new LCNP4 board has the correct LCN pin assignments.
8	Insert the new LCNP4 board assembly firmly into in the PCI5 slot connector until the assembly card is fully seated.
9	Install the screw that secures the assembly card bracket to the back panel.
10	Close the card-guide latch.
11	Replace the cover and bezel.
12	Reconnect the system to its electrical outlet and turn the system on, including any attached peripherals.
13	Replace the PE1600 in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".
14	Press the Power On/Off button on the front of the platform to turn the power back on.
15	Run the Configurator to re-establish the correct node number.
16	Reconnect the LCN MAU cable.
17	Restart the Operating System.

4.3 Servicing Hard Disk Drives/Power Supply (PE1600)

Overview

The PE1600SC is configured with a pair of redundant hard disk drives and redundant power supplies. Both the hard disk drives and power supplies are hot swappable. You must, however, remove and replace only one power supply or hard disk drive at a time in a system that is powered on. Refer to the Dell documentation for detailed instructions on swapping the power supply and hard disk drive.

References

The following table lists the specific Dell publications and relevant sections for troubleshooting and servicing the hard disk drives and power supplies.

Table 4-1 Dell PE1600 Hard Disk Drives/Power Supply Publications

Publication	Contains these sections
<i>Dell™ PowerEdge™ 1600SC Systems Service Manual</i>	System Overview Basic Troubleshooting Indicators, Codes, and Messages Removing and Installing Parts Jumpers and Connectors Using the System Setup Program
<i>Dell™ PowerEdge™ 1600SC Systems Installation and Troubleshooting Guide</i>	Introduction Indicators, Codes, and Messages Finding Software Solutions Running the System Diagnostics Troubleshooting Your System Installing System Options Installing Drives Getting Help Jumpers, Switches, and Connectors I/O Ports and Connectors Abbreviations and Acronyms

4. Platform Servicing

4.4. Servicing Hard Disk Drives/Power Supply (PE1800)

4.4 Servicing Hard Disk Drives/Power Supply (PE1800)

Overview

The PE1800 may be configured with a pair of redundant hard disk drives and redundant power supplies. Both the hard disk drives and power supplies are hot swappable. You must, however, remove and replace only one power supply or hard disk drive at a time in a system that is powered on. Refer to the Dell documentation for detailed instructions on swapping the power supply and hard disk drive.

References

The following table lists the specific Dell publications and relevant sections for troubleshooting and servicing the hard disk drives and power supplies.

Table 4-2 Dell PE1800 Hard Disk Drives/Power Supply Publications

Publication	Contains these sections
<i>Dell™ PowerEdge™ 1800 Systems Service-Only Parts Replacement Procedures</i>	Before You Begin Recommended Tools Nonredundant Power Supply Power Distribution Board Control Panel SCSI Backplane System Board
<i>Dell™ PowerEdge™ 1800 Systems Installation and Troubleshooting Guide</i>	Introduction Indicators, Codes, and Messages Finding Software Solutions Running System Diagnostics Troubleshooting Your System Installing System Options Installing Drives Getting Help Jumpers, Switches, and Connectors I/O Ports and Connectors

4.5 Servicing Honeywell Options in PE1600/PE1800

Overview



CAUTION

Be careful not to damage the EMI gasket fingers when removing/installing boards.



ESD HAZARD

Use a grounding strap and grounded work surfaces and equipment when handling any electrostatically sensitive components such as the video cards, NIC adapter cards, and SCSI controller cards. Store and transport parts only in electrostatically safe containers.

Replacing cards in the expansion slots

Use the following procedure to replace the expansion cards in the PCI slots. See also your platform service manual.



ESD HAZARD

Expansion cards are electrostatically sensitive device. Use a grounding strap and grounded work surfaces and equipment when handling these components. Store and transport parts only in electrostatically safe containers.

4. Platform Servicing

4.5. Servicing Honeywell Options in PE1600/PE1800

Step	Action
1	Perform all necessary procedures in Section 4.1, "Servicing the PE1600/PE1800 Enclosure" to access the LCNP4 assembly., including: • Shutting down system • Disconnecting power from the computer • Accessing the inside of the electronics enclosure Disconnecting cables from the card being replaced.
2	Turn off the system, including any attached peripherals, and disconnect the system from the electrical outlet.
3	Remove the bezel and the cover using the "Accessing the electronics" procedure in Section 4.1.
4	Disconnect cables from the expansion card
5	Remove the filler bracket from the expansion slot.
6	While wearing a grounded ESD wrist strap, remove the existing card from the PCI slot.
7	Insert the new board firmly into in the PCI slot connector until the assembly card is fully seated.
8	Install the screw that secures the assembly card bracket to the back panel.
9	Close the card-guide latch.
10	Replace the cover and bezel.
11	Reconnect the cable to the card.
12	Reconnect the system to its electrical outlet and turn the system on, including any attached peripherals.
13	Replace the PE1600 in the furniture using the appropriate installation procedures in Section 2, "Platform Installation".
14	Press the Power On/Off button on the front of the platform to turn the power back on.
15	Restart the Operating System.

